W2 – Ratios for Angles Greater than 90°

MCR3U

Jensen

1) Sketch each angle in standard position and determine the related acute angle.

a) 135^0 b) 210^0 c) 315^0 d) -30^0

e) -225^0 f) -330^0 g) 150^0 h) -120^0

i) 105^0 j) -163^0 k) -141^0 l) -280^0

2) State the value of each ratio **exactly**.

a) sin 225° **c)** sin 270°

b) cos 240° **d)** tan 300°

e) cos 180°

3) Point P (-9, 4) is on the terminal arm of an angle in standard position.

a) Sketch the principal angle, θ.

b) What is the measure of β, the related acute angle to the nearest degree?

c) What is the measure of θ to the nearest degree?

4) Point P (7, -24) is on the terminal arm of an angle in standard position.

a) Sketch the principal angle, θ.

b) What is the measure of β, the related acute angle to the nearest degree?

c) What is the measure of θ to the nearest degree?

Answers

1) a) 45^0 b) 30^0 c) 45^0 d) 30^0 e) 45^0 f) 30^0 g) 30^0 h) 60^0 i) 75^0 j) 17^0 k) 39^0 l) 80^0

2) a) $-\frac{1}{\sqrt{2}}$ b) $-\frac{1}{2}$ c) -1 d) $-\sqrt{3}$ e) -1

3) b) 24^0 c) 156^0

4) b) 74^0 c) 286^0